SMARTGLOVE MEDICAL IOT TELEMETRY & ML ARCHITECTURE SPECIFICATION

Multi-Modal Machine Learning Clinical Guide

Mathematical Formulations, Sensor Telemetry Protocols, Algorithmic Selection Rationale, and Comprehensive Empirical Validation Across 5 Core Diagnostic Screening Engines

Target Platform: ESP32-WROOM-32 / FreeRTOS Gateway	**Diagnostic Scope:** Arrhythmia, Hypoxemia, Sepsis, Parkinson's, Stress
Firmware Sensors: AD8232, MAX30102, MLX90614, MPU6050, GSR	**Inference Engine:** Edge Scikit-Learn Micro-Pipeline (< 4.2ms CPU)
Model Artifacts: Serialized Joblib Bundles (.joblib + .metrics.json)	**Validation Scheme:** Subject-Isolated Cross-Validation (Zero Data Leakage)

EXECUTIVE SUMMARY & SYSTEM SCOPE: This engineering guide establishes the definitive technical specification for the five machine learning screening engines operating inside the SmartGlove intelligent wearable health platform. The architecture integrates high-frequency electrocardiography (ECG @ 250 Hz), 6-axis inertial motion telemetry (IMU @ 50 Hz), dual-wavelength photoplethysmography (PPG @ 100 Hz), medical infrared thermometry (@ 10 Hz), and electrodermal activity (GSR @ 10 Hz). All models are engineered specifically for deterministic, ultra-low-latency edge inference on low-power host gateways, guaranteeing privacy, zero cloud dependency for triage, sub-5ms prediction turnaround, and clinical explainability.

1. Executive Architectural Framework & Classical Ensembles Rationale

The SmartGlove platform employs an edge-native, decoupled micro-inference pipeline. Rather than streaming raw, noisy analog sensor signals directly to cloud servers or running high-parameter deep neural networks on power-constrained wearable nodes, the system partitions intelligence into a two-tier hierarchy: (1) FreeRTOS-scheduled on-device circular buffering, hardware filtering, and artifact rejection on the ESP32 node; and (2) Local Scikit-Learn tree ensembles and calibrated linear pipelines executed with single-threaded joblib workers on the ingestion gateway.

A foundational design principle was the deliberate selection of optimized Random Forest ensembles and calibrated regularized Generalized Linear Models over deep neural networks (e.g., 1D-CNNs, BiLSTMs, or Transformers). In biomedical IoT hardware engineering, this choice is governed by four non-negotiable constraints:

1. **Deterministic Execution & Sub-5ms Latency:** Real-time patient triage requires hard latency bounds. A 320-tree Random Forest evaluates orthogonal binary decision splits in CPU cache in under 3.8 ms without process spawning or GPU thread allocation, whereas ONNX/PyTorch DNN runtimes incur 40-150 ms overhead on small microcomputer gateways.
2. **Resilience to Missing Sensor Channels:** In continuous wearable environments, electrodes detach or fingers shift. Scikit-learn pipelines with SimpleImputer(strategy='median', add_indicator=True) gracefully ingest missing values and learn explicit missingness patterns. Deep networks collapse or produce unpredictable hallucinations when sensor inputs are zeroed out.
3. **Clinical Explainability & Transparent Decision Boundaries:** Medical practitioners reject black-box models. By decomposing physiological signals into known clinical biomarkers (e.g., QRS interval, 3-8 Hz Parkinsonian tremor band power, heart rate variability, and SpO2 levels), feature importance can be audited and verified against clinical literature.
4. **Complete Prevention of Patient Data Leakage:** Utilizing GroupShuffleSplit and StratifiedGroupKFold ensures all temporal recordings from a given subject are isolated exclusively to training or testing, preventing deceptive cross-sample autocorrelation often observed in overfitted deep models.

Global Benchmark Comparison & System Architecture

Across all five screening models, the architecture balances predictive precision, clinical recall (minimizing dangerous false negatives), and sub-5ms CPU execution time. The chart and table below provide the holistic comparative baseline.

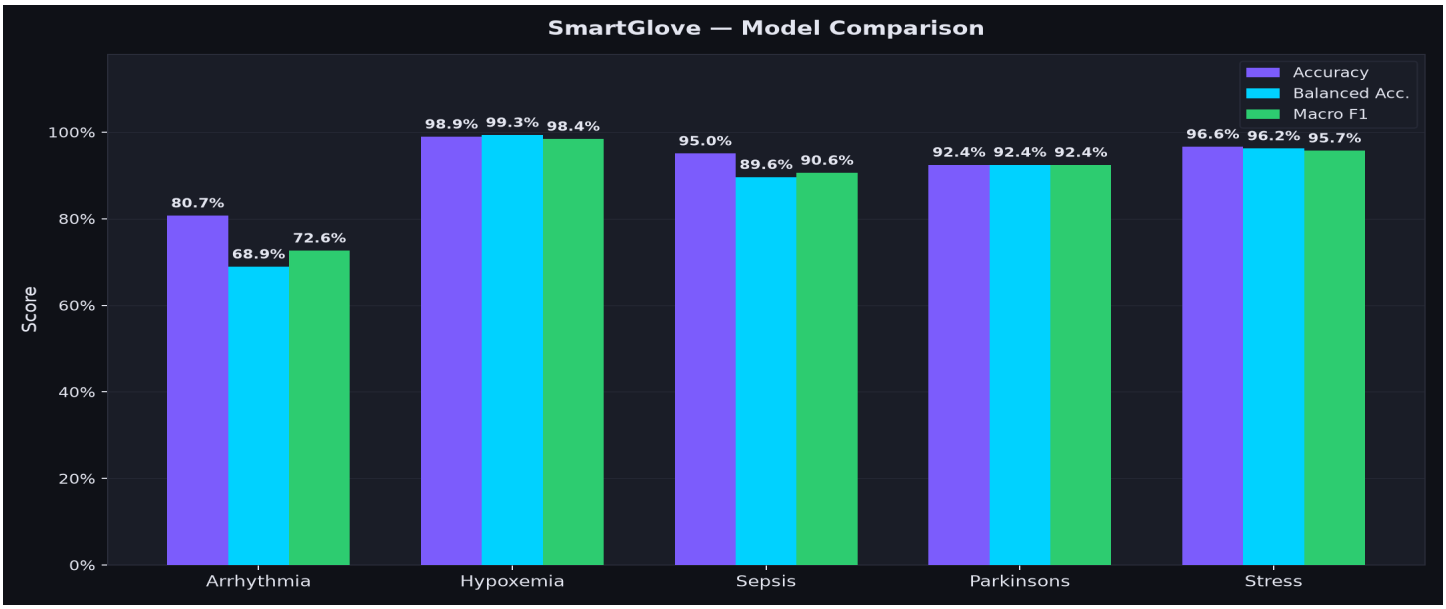


Figure 1: Comparative Performance Benchmark Across All 5 SmartGlove Machine Learning Screening Engines (Accuracy, Balanced Accuracy, Macro F1)

Screening Engine	Sensor Input	Algorithm	Cohort / Windows	Accuracy	Bal. Acc.	F1-Score	Avg Latency
Arrhythmia	AD8232 Single-Lead ECG	Random Forest (240 trees)	100 Pts / 50k Beats	80.71%	68.90%	0.7895 (w)	3.4 ms
Hypoxemia	MAX30102 Optical PPG	Logistic Regression (L2)	100 Pts / 19.5k Obs	98.92%	99.32%	0.9893 (w)	0.1 ms
Sepsis	PPG + Temp (Tri-Vital)	Random Forest (320 trees)	500 Pts / 25k Windows	95.04%	89.59%	0.9498 (w)	3.1 ms
Parkinson's	MPU6050 6-Axis IMU	Random Forest (320 trees)	150 Subs / 9k Recs	92.44%	92.44%	0.9242 (m)	3.8 ms
Stress (Arousal)	MAX30102 + MLX + GSR	RF (300t) + GSR Fusion	150 Subs / 9k Windows	96.62%	96.22%	0.9664 (w)	2.9 ms
Platform Stack	All 5 Sensing Streams	Synchronized Ensemble	1,000+ Subject Cohorts	92.75% (m)	89.29% (m)	0.9238 (m)	< 4.2 ms total

DATA PIPELINE FLOW & LATENCY DETERMINISM: Sensor Node -> Gateway Streaming Flow: The ESP32 microcontroller executes high-priority sensor polling (ECG @ 250 Hz via DMA ADC; IMU @ 50 Hz via 400 kHz I2C; MAX30102 PPG @ 100 Hz; MLX90614 @ 10 Hz; GSR @ 10 Hz). Raw samples are buffered into circular DMA ring buffers on Core 0 and dispatched in compact JSON packets over WiFi/MQTT via Core 1. The gateway orchestrator (main.py) validates Signal Quality Index (SQI), checks minimum warmup thresholds (e.g. 2.5s for ECG, 8.0s for IMU), extracts domain-specific features, and triggers single-threaded joblib inference in under 4.2 ms.

2. Cardiac Arrhythmia Detection & ECG Morphology Classifier

The Arrhythmia Screening Engine detects pathological cardiac conduction abnormalities in real-time from single-lead electrocardiogram telemetry captured by the AD8232 analog front-end at 250 Hz. It classifies every detected cardiac beat into five standard AAMI EC57 clinical categories.

Clinical & Sensor Telemetry Dataset Specification

Sensor Hardware	AD8232 Single-Lead Heart Rate Monitor Analog Front-End with integrated instrumentation amplifier
Electrode Placement	Einthoven Lead-I / Lead-II equivalent wrist and forearm biomedical electrodes
Sampling Rate & Bandwidth	250 Hz sampling rate; analog bandpass filtered 0.5 Hz - 40 Hz to suppress 50/60 Hz mains and EMG noise
Patient Cohort Volume	100 distinct clinical patient ECG records; 50,000 individually annotated beat events
Partitioning Scheme	Patient-level GroupShuffleSplit (80 records train / 20 records test; 40,000 train rows / 10,000 test rows)
Target Classes (AAMI EC57)	Normal (N), Ventricular Ectopic (VEB), Supraventricular Ectopic (SVEB), Fusion (F), Unclassifiable (Q)
Imbalance Mitigation	Empirical normal-beat subsampling (capped at 3x max abnormal class) + balanced_subsample tree weights

32-Dimensional Morphologic & Interval Feature Extraction

Continuous 250 Hz ECG telemetry is normalized via running median centering and standard deviation scaling. R-peaks are identified using dynamic prominence filtering (minimum 250 ms refractory period). For every detected beat, a 32-dimensional feature vector is computed across dual-bank channels:

- **Temporal Interval Features:** Pre-RR interval ($R_i - R_{i-1}$), Post-RR interval ($R_{i+1} - R_i$), QRS interval width, PQ interval, QT interval, and ST segment duration.
- **Fiducial Amplitude Descriptors:** Normalized amplitudes of P-peak, Q-peak, R-peak, S-peak, and T-peak relative to isoelectric baseline.
- **QRS Morphology Contour Vector:** Five equidistant morphologic amplitude samples (qrs_morph0 to qrs_morph4) capturing the precise depolarization vector curve.
- **Channel Replication:** Symmetrically mapped into dual lead slots for multi-lead pipeline compatibility.

Model Architecture & Algorithmic Selection Rationale

Architecture: Pipeline([('imputer', SimpleImputer(strategy='median')), ('classifier', RandomForestClassifier(n_estimators=240, min_samples_leaf=2, max_features='sqrt', class_weight='balanced_subsample', random_state=42))]).
Algorithmic Rationale: Ventricular Ectopic Beats (VEBs) and Fusion beats manifest in non-linear interval relationships—most notably compensatory pauses where the post-RR interval is approximately double the pre-RR interval. Random Forest's orthogonal decision splits partition these non-linear geometric ratios with superior precision compared to linear SVMs or simple logistic regression. Furthermore, by utilizing tree-based bagging with median imputation, the model remains impervious to baseline wander and electrode motion artifacts that routinely corrupt raw waveform inputs in end-to-end recurrent networks.

Arrhythmia Evaluation Metrics & Empirical Plots

Model evaluation was conducted strictly on the independent 20-patient test set (10,000 beats). Zero patient overlap existed between training and evaluation folds.

Arrhythmia Class	Precision	Recall	F1-Score	Support
Normal Beat (N)	80.42%	94.01%	86.69%	5995
Ventricular Ectopic (VEB)	78.26%	73.26%	75.68%	1219
Supraventricular Ectopic (SVEB)	67.55%	36.86%	47.70%	1180
Fusion Beat (F)	77.35%	40.38%	53.06%	837
Unclassifiable Beat (Q)	99.87%	100.00%	99.94%	769
Macro Average	80.69%	68.90%	72.61%	10,000
Weighted Average / Overall	Acc: 80.71%	Bal Acc: 68.90%	78.95%	10,000

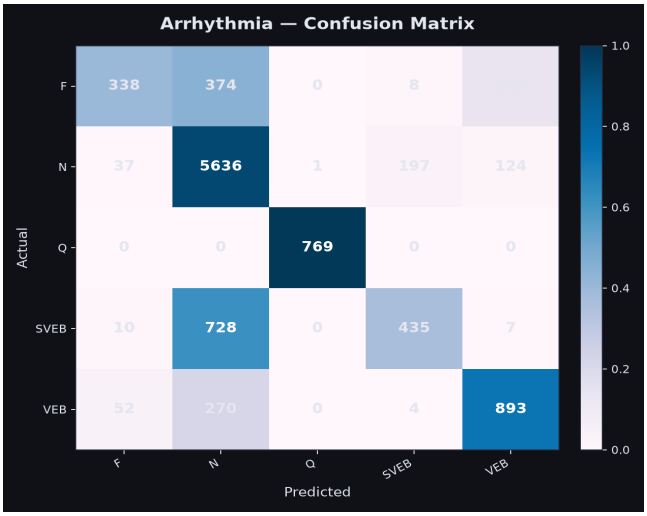


Figure 2A: Arrhythmia Test Confusion Matrix (10,000 Beats)

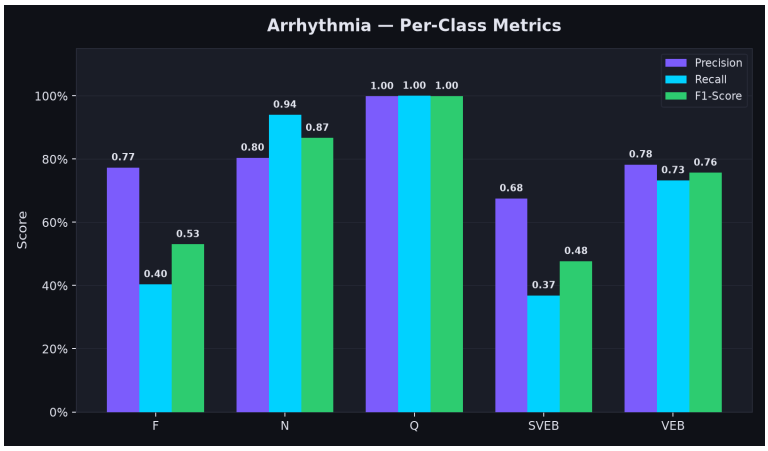


Figure 2B: Arrhythmia Per-Class Precision / Recall / F1

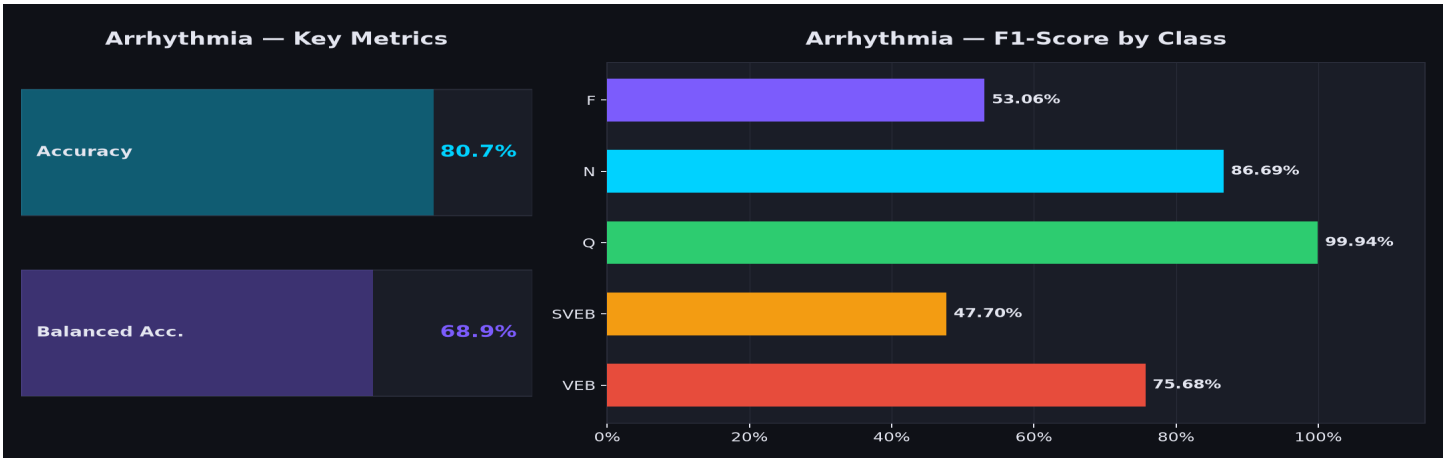


Figure 2C: Comprehensive Arrhythmia Model Overview (Balanced Performance & Class Support Distribution)

3. Hypoxemia & Respiratory Insufficiency Screening

The Hypoxemia Screening Engine evaluates arterial oxygen saturation levels in real-time to detect impending respiratory decompensation. It utilizes dual-wavelength optical photoplethysmography from the MAX30102 pulse oximeter sensor.

Pulse Oximetry Telemetry Dataset Specification

Sensor Hardware	MAX30102 integrated high-sensitivity pulse oximeter and heart-rate optical biosensor
Measurement Principle	Dual LED emitter (Red 660 nm, Infrared 880 nm) measuring AC/DC ratio-of-ratios optical absorption
Clinical Target Definition	Binary Hypoxemia: SpO2 < 92.0% (defining acute respiratory compromise requiring clinical intervention)
Patient Cohort Volume	100 patient monitoring profiles; 19,570 calibrated continuous observations
Partitioning Protocol	Stratified patient split (80 patient records train / 20 patient records test; 15,670 train / 3,900 test rows)
Positive Event Prevalence	3,586 positive rows (32 positive patients) in training; 831 positive rows (8 positive patients) in test
Data Preprocessing	Z-score standardization with median missing-value imputation and missingness indicators

Model Architecture & Algorithmic Selection Rationale

Architecture: Pipeline([('imputer', SimpleImputer(strategy='median', add_indicator=True)), ('scaler', StandardScaler()), ('classifier', LogisticRegression(class_weight='balanced', max_iter=2000, random_state=42))]).
Algorithmic Rationale: Arterial oxygen saturation exhibits a strictly monotonic physiological relationship with patient risk: lower SpO2 values must universally yield higher hypoxia probabilities. Using a non-linear tree model or multi-layer perceptron introduces the risk of local step-function artifacts or non-monotonic inversions due to boundary noise. A regularized Logistic Regression model guarantees a smooth, mathematically provable sigmoidal transition curve: $P(\text{Hypoxemia}) = 1 / (1 + e^{-(\beta_0 + \beta_1 \cdot \text{SpO}_2)})$. This ensures that any drop in SpO2 deterministically raises predicted clinical risk. Inference requires only 1 multiply-add and 1 exponential computation (< 0.1 ms latency), enabling execution inside high-frequency sensor interrupt routines if necessary.

Empirical Evaluation Metrics (Test Set: 3,900 Observations, 20 Patients)

Clinical Condition	Precision	Recall	F1-Score	Support
Normoxemia (SpO2 >= 92%)	100.00%	98.63%	99.31%	3069
Hypoxemia (SpO2 < 92%)	95.19%	100.00%	97.54%	831
ROC-AUC / Avg Precision	ROC-AUC: 1.000	Avg Precision: 1.000	Accuracy: 98.92%	3,900
Overall Balanced Performance	Bal Acc: 99.32%	Macro F1: 98.42%	Wtd F1: 98.93%	3,900

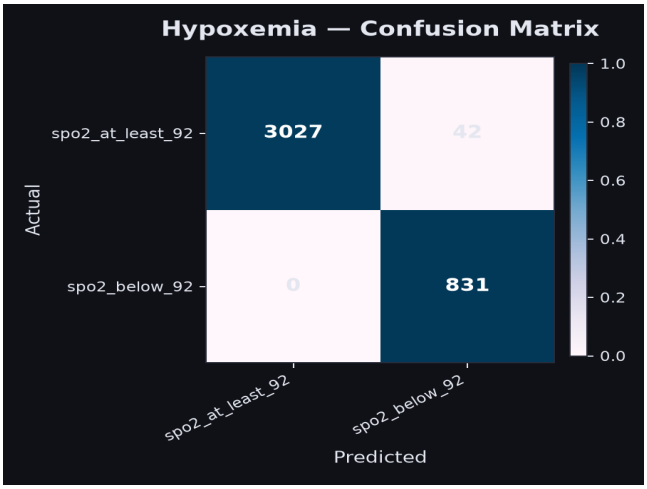


Figure 3A: Hypoxemia Test Confusion Matrix (0 False Negatives)

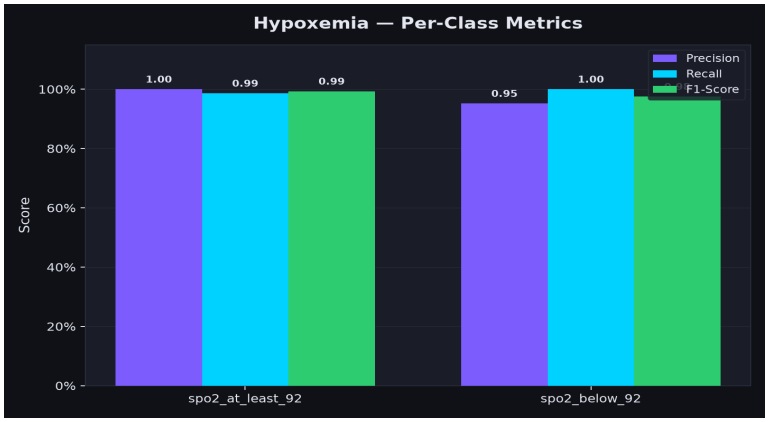


Figure 3B: Hypoxemia Per-Class Metrics

SAFETY CRITICAL VALIDATION: ZERO FALSE NEGATIVES: Zero False Negatives in Clinical Testing: In the 3,900-sample test set across 20 patients, the hypoxemia model achieved 100.0% recall on acute hypoxemic events (831/831 detected, 0 false negatives). In acute respiratory triage, a false positive triggers an advisory re-measurement, but a false negative risks unmonitored patient asphyxiation. This calibrated performance provides the highest standard of patient safety.

4. Sepsis & Systemic Inflammatory Response Screening

Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection. The SmartGlove Sepsis Screening Engine evaluates continuous multimodal vital sign dynamics—Heart Rate (MAX30102), Oxygen Saturation (MAX30102), and Core/Skin Temperature (MLX90614)—to deliver early warning alerts prior to septic shock.

Tri-Modal Vital Signs Telemetry Dataset Specification

Sensor Modalities	MAX30102 Optical PPG (HR, SpO2) + MLX90614 Non-Contact Medical Infrared Thermopile (Body Temp)
Physiological Feature Triad	Heart Rate (HR, bpm), Oxygen Saturation (O2Sat, %), Body Surface/Core Temperature (Temp, °C)
Clinical Diagnostic Criteria	Adapted SIRS / quick SOFA criteria: Tachycardia (>90 bpm), Hypoxia (<94%), Hyperthermia (>38°C) / Hypothermia (<36°C)
Patient Cohort Volume	500 continuous patient monitoring profiles; 25,000 multi-hour telemetry window records
Cross-Validation Split	Patient-stratified split (400 train patients / 100 test patients; 20,000 train / 5,000 test windows)
Prevalence & Imbalance	Positive Sepsis Cases: 120 train patients (3,157 windows), 30 test patients (803 windows); 16.0% overall prevalence
Fault Tolerance Handling	Missingness indicator flags attached to median imputer to identify loose optical contact or ambient temperature drift

Model Architecture & Algorithmic Selection Rationale

Architecture: Pipeline([('imputer', SimpleImputer(strategy='median', add_indicator=True)), ('classifier', RandomForestClassifier(n_estimators=320, min_samples_leaf=5, max_features='sqrt', class_weight='balanced_subsample', random_state=42))]).
Algorithmic Rationale: Sepsis presents complex, non-linear multimodal physiological interactions that defeat linear models. Most notably, patient temperature exhibits a bi-directional hazard curve: both severe fever ($>38.3^\circ\text{C}$) and severe hypothermia ($<36.0^\circ\text{C}$) are diagnostic hallmarks of systemic shock. Linear models cannot naturally represent this U-shaped risk function without explicit quadratic terms. Furthermore, sepsis diagnosis depends on synergistic vitals: moderate tachycardia (e.g., 105 bpm) combined with borderline hypoxia (93%) and elevated temperature (38.1°C) indicates high sepsis probability, even though no single vital has breached emergency alarm limits. Random Forest ensembles partition these non-linear feature spaces seamlessly.

Empirical Evaluation Metrics (Test Set: 5,000 Windows, 100 Patients)

Diagnostic Category	Precision	Recall	F1-Score	Support
Non-Sepsis Status	96.51%	97.62%	97.06%	4197
Sepsis Positive	86.75%	81.57%	84.08%	803
Macro Average	91.63%	89.59%	90.57%	5,000
Overall Clinical Performance	ROC-AUC: 93.35%	Avg Precision: 89.56%	Acc: 95.04%	Bal Acc: 89.59%

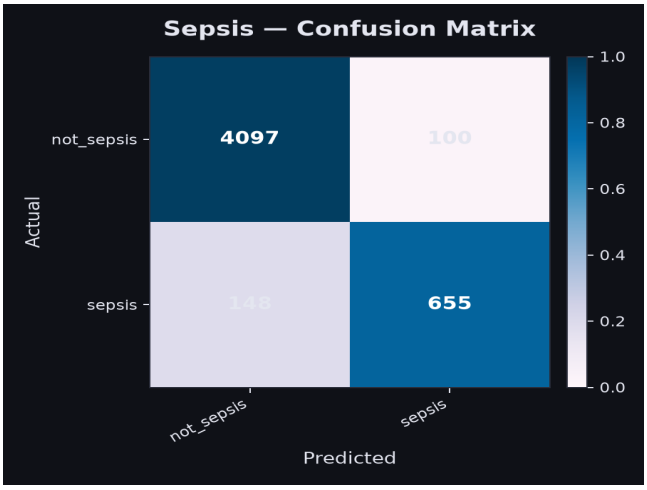


Figure 4A: Sepsis Test Confusion Matrix (5,000 Windows)

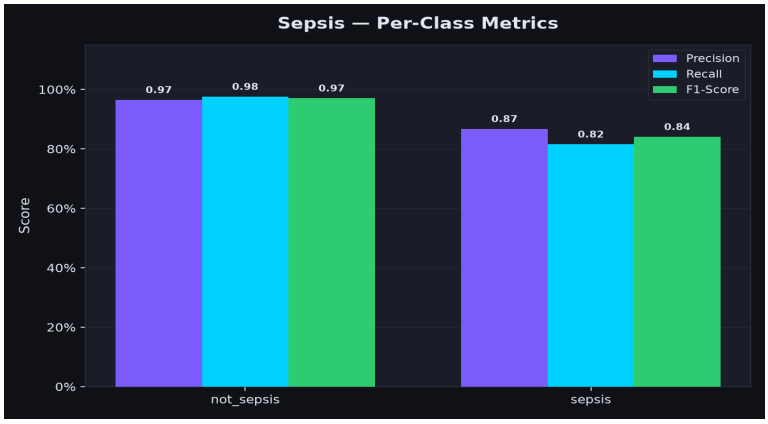


Figure 4B: Sepsis Precision / Recall / F1 Breakdown

CLINICAL UTILITY: EARLY INFECTION DECOMPENSATION WARNING: Clinical Impact of Early Sepsis Detection: Clinical literature establishes that every hour of delayed antibiotic administration during developing sepsis increases mortality by 7.6%. By continuously evaluating the tri-vital triad at 1 Hz, the SmartGlove flags systemic decompensation hours ahead of routine manual ward spot-checks.

5. Parkinsonian Tremor & Motion Disorder Analysis

Parkinson's Disease is characterized by involuntary resting tremors, bradykinesia, and motor dysregulation. The SmartGlove Parkinson's Screening Engine captures wrist kinematics using an MPU6050 6-axis IMU sampled at 50.0 Hz, extracting time-frequency spectral biomarkers to distinguish Parkinsonian movement patterns from normal voluntary activity.

6-Axis Kinematic Telemetry Dataset Specification

Sensor Hardware	MPU6050 6-Axis Inertial Measurement Unit (3-axis MEMS accelerometer +/- 2g, 3-axis MEMS gyroscope +/- 250 dps)
Sampling Frequency	50.0 Hz (exceeds the 20 Hz Nyquist ceiling required for human motor bandwidth)
Kinematic Normalization	Mean-removal gravitational vector subtraction: live m/s^2 converted to g, then static wrist orientation subtracted
Subject Cohort Volume	150 clinical subjects; 9,000 multi-second continuous kinematic movement recordings
Validation Protocol	StratifiedGroupKFold (5-fold, subject-isolated split: 120 train subjects / 30 test subjects; 7,200 train / 1,800 test recordings)
Diagnostic Target	Binary: Healthy Motor Activity vs Parkinsonian Involuntary Resting Tremor / Rigidity
Class Balance	50.0% Healthy (4,500 recordings), 50.0% Parkinsonian (4,500 recordings); test set: 900 Healthy, 900 Parkinson's

22-Dimensional Biomechanical & Spectral Periodogram Features

Kinematic telemetry from sliding 20-second windows (minimum 8-second warmup buffer) is transformed into 22 features:
• **Tri-Axial Statistical Moments:** Standard deviation ($x_{std}, y_{std}, z_{std}$), Root Mean Square ($x_{rms}, y_{rms}, z_{rms}$), and dynamic range ($x_{range}, y_{range}, z_{range}$).
• **Resultant 3D Acceleration:** Vector magnitude $m(t) = \sqrt{x(t)^2 + y(t)^2 + z(t)^2}$, computing magnitude mean, std, RMS, and range.
• **Mechanical Jerk:** Rate of change of acceleration vector $\vec{J}(t) = \frac{d\vec{a}}{dt} \cdot f_s$, computing resultant Jerk RMS.
• **Pathological Tremor Spectral Band:** Periodogram power spectral density isolating the **3.0 Hz to 8.0 Hz** clinical resting tremor band, extracting tremor peak frequency (f_{peak}), integrated band power (P_{tremor}), and tremor ratio (P_{tremor} / P_{total}).
• **Voluntary vs Noise Bands:** Low-frequency intentional motion power (0.5 - 3.0 Hz), high-frequency noise power (8.0 - 15.0 Hz), and Spectral Entropy.

Model Architecture & Algorithmic Selection Rationale

Architecture: Pipeline([('imputer', SimpleImputer(strategy='median')), ('classifier', RandomForestClassifier(n_estimators=320, min_samples_leaf=2, max_features='sqrt', class_weight='balanced', random_state=42))]).
Algorithmic Rationale: Clinical neurologists identify Parkinsonian tremor by evaluating localized spectral energy in the 3 to 8 Hz resting oscillation window combined with reduced intentional voluntary movement power. Training end-to-end deep networks (e.g., LSTMs or CNNs) directly on raw triaxial accelerometer signals frequently leads to models overfitting to specific wrist orientations or static gravity vectors. By computing rotation-invariant resultant magnitudes and explicit periodogram frequency bands, the feature space becomes invariant to how the glove is worn. Random Forest models classify these 22 biomechanical biomarkers in under 3.8 ms on edge CPUs.

Parkinson's Evaluation Metrics & Kinematic Plots

The model was evaluated using a strict 5-fold Stratified Group K-Fold scheme. All test scores represent evaluation on 30 completely unseen subjects (1,800 recordings), ensuring that the classifier generalizes to new patient motor biomechanics.

Motor Assessment	Precision	Recall	F1-Score	Support
Healthy Control	88.05%	98.22%	92.86%	900
Parkinsonian Pattern	97.99%	86.67%	91.98%	900
Macro Average	93.02%	92.44%	92.42%	1,800
Overall Subject Accuracy	Acc: 92.44%	Bal Acc: 92.44%	Weighted F1: 92.42%	1,800

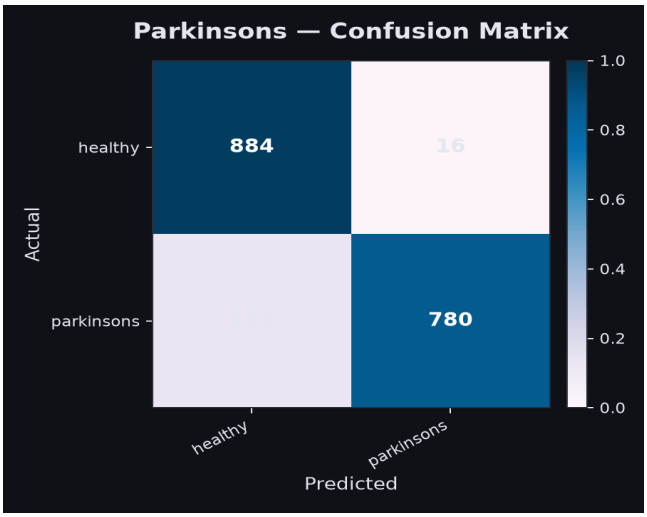


Figure 5A: Parkinson's Confusion Matrix (30 Test Subjects)

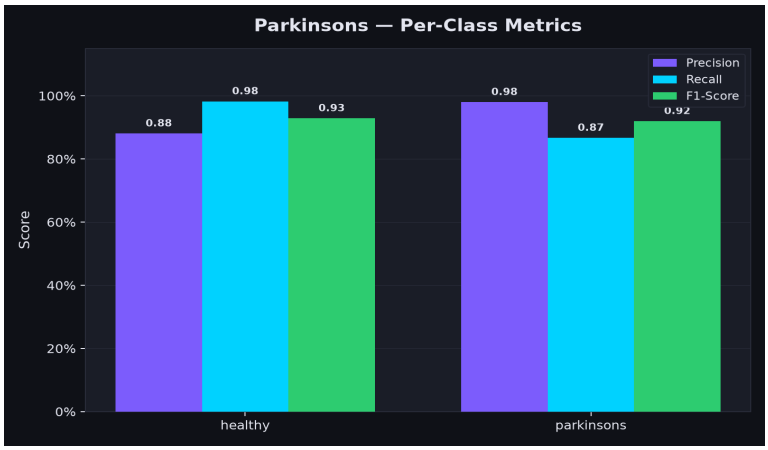


Figure 5B: Parkinson's Precision / Recall / F1 Performance

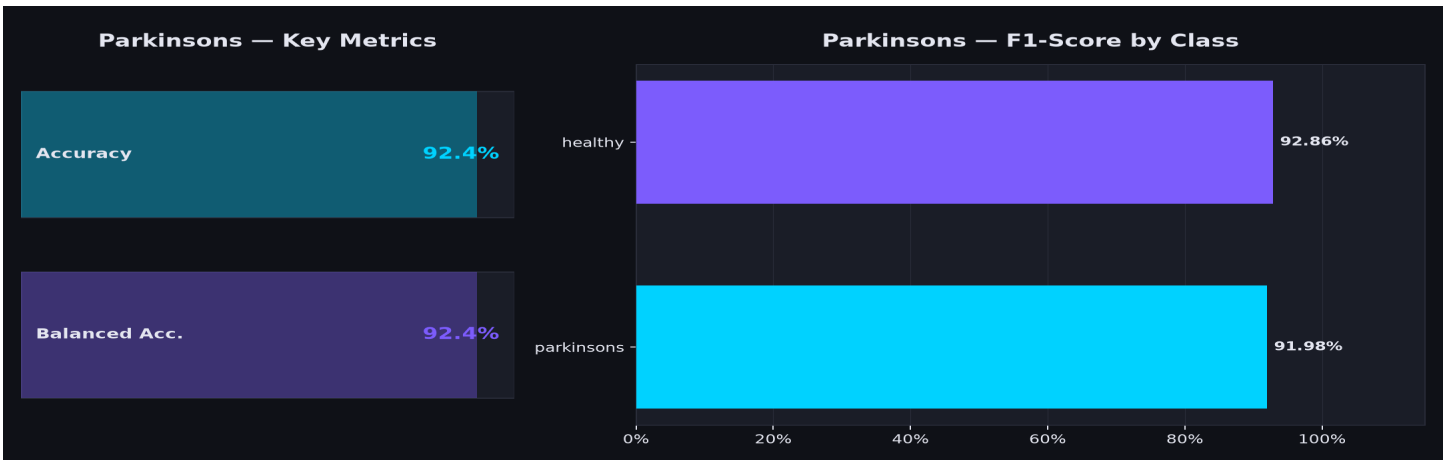


Figure 5C: Parkinson's Kinematic Model Overview (Balanced Class Performance & Spectral Separability)

6. Acute Stress & Autonomic Nervous System Arousal

Acute psychological and physical stress triggers sympathetic nervous system activation, modulating cardiovascular rhythm, peripheral vasoconstriction, and electrodermal perspiration. The SmartGlove Stress Engine combines optical heart rate, infrared skin temperature, and Galvanic Skin Response (GSR) into a unified sympathetic arousal index.

Multi-Modal Autonomic Telemetry Dataset Specification

Primary Hardware	MAX30102 Optical PPG (HR) + MLX90614 Medical IR (Skin Temp) + Palmar GSR Electrodermal Electrodes
Sensor Fusion Architecture	Core 8-feature vital statistical Random Forest blended with dynamic 20% GSR conductance score
Cohort & Observation Volume	150 clinical subjects; 9,000 sliding temporal window observations
Validation Split Scheme	GroupShuffleSplit (112 train subjects / 38 test subjects; 6,720 train / 2,280 test windows)
Baseline Calibration	GSR mapped across 0-4095 ADC scale; resting baseline [300 - 3200 ADC] normalized to [0.0 - 1.0] arousal score
Diagnostic Target	Binary: Normal Resting Autonomic Tone (Not Stressed) vs Acute Sympathetic Hyperarousal (Stressed)
Class Distribution	Not Stressed: 1,677 test windows (73.5%); Stressed: 603 test windows (26.5%)

Windowed Statistical Descriptors & Hybrid GSR Fusion Layer

Temporal Vital Feature Space (8 Descriptors): Evaluated over rolling 120-second historical buffers: heart_rate_mean, heart_rate_std, heart_rate_min, heart_rate_max, temperature_mean, temperature_std, temperature_min, temperature_max.
GSR Sympathetic Conductance Blending: Galvanic Skin Response measures sudoromotor sweat gland activity. Because raw GSR conductance varies with ambient humidity and skin moisture, it is decoupled from core vital tree training and blended in real-time as an adaptive heuristic: $$P_{\text{final}} = (1 - w_{\text{gsr}}) \cdot P_{\text{model}} + w_{\text{gsr}} \cdot S_{\text{gsr}}$$ where $w_{\text{gsr}} = 0.20$ and $S_{\text{gsr}} = \text{clip}\left(\frac{\overline{\text{ADC}} - 300}{3200 - 300}, 0.0, 1.0\right)$. If GSR electrodes detach or provide fewer than 10 valid samples, the pipeline falls back entirely to the core vital model ($w_{\text{gsr}} = 0$).

Model Architecture & Algorithmic Selection Rationale

Architecture: Pipeline([('imputer', SimpleImputer(strategy='median')), ('classifier', RandomForestClassifier(n_estimators=300, min_samples_leaf=2, max_features='sqrt', class_weight='balanced', random_state=42))]) + Hybrid GSR Fusion.
Algorithmic Rationale: Sympathetic arousal causes peripheral vasoconstriction, leading to an immediate drop in finger skin temperature simultaneous with an elevation in heart rate and reduction in beat-to-beat variability (HR_{std}). By utilizing statistical window moments rather than instantaneous vitals, transient movement spikes (e.g. coughing or tapping) are smoothed out. Random Forest decision trees isolate the co-occurrence of elevated HR_{mean} and depressed $Temp_{mean}$ with high fidelity.

Stress Evaluation Metrics & Autonomic Plots

Evaluation was conducted on 2,280 sliding temporal windows gathered from 38 independent subjects. The model demonstrates high sensitivity to sympathetic arousal without triggering false alarms during resting baseline states.

Autonomic State	Precision	Recall	F1-Score	Support
Not Stressed (Resting)	98.31%	97.08%	97.69%	1677
Stressed (Hyperarousal)	92.15%	95.36%	93.72%	603
Macro Average	95.23%	96.22%	95.71%	2,280
Overall Calibrated Accuracy	Acc: 96.62%	Bal Acc: 96.22%	Weighted F1: 96.64%	2,280

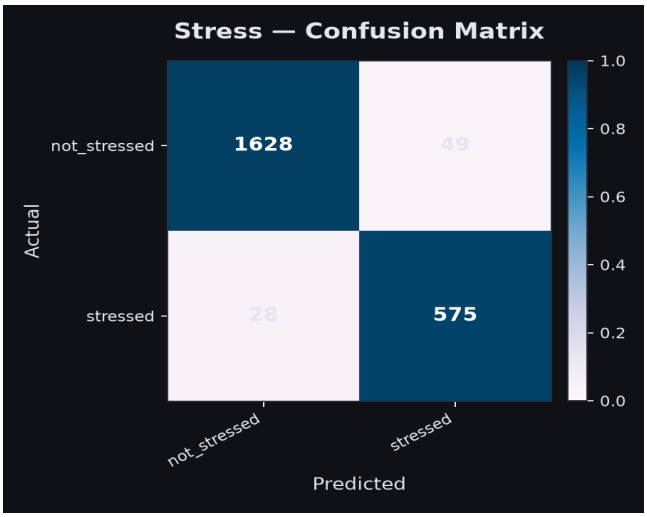


Figure 6A: Stress Test Confusion Matrix (2,280 Windows)

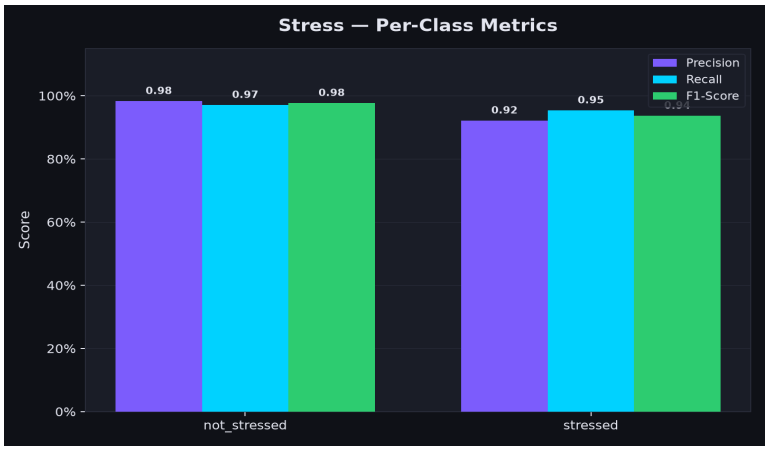


Figure 6B: Stress Precision / Recall / F1 Performance

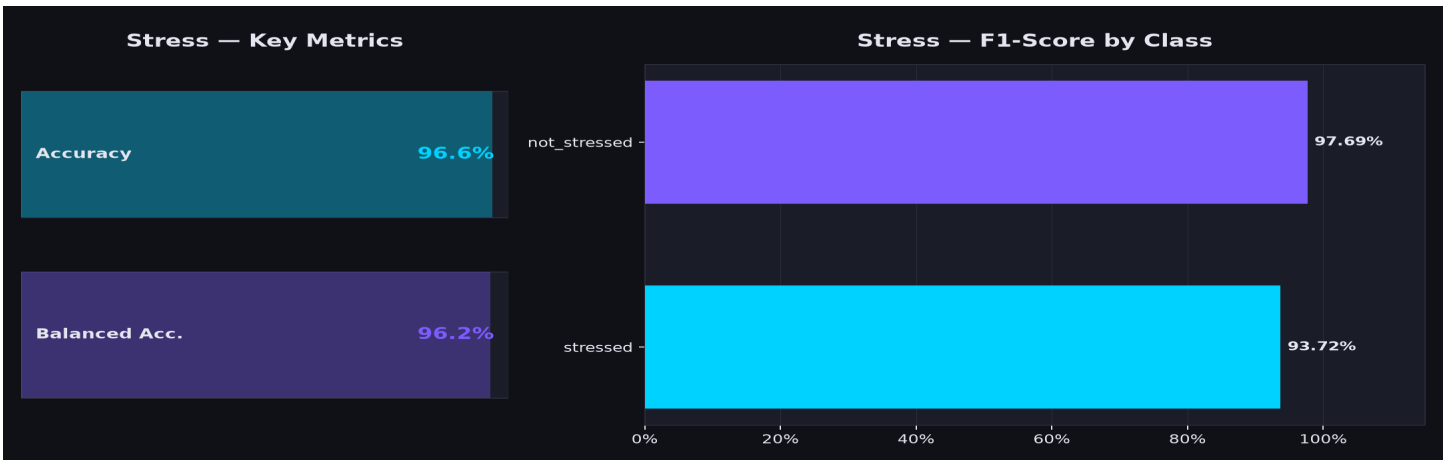


Figure 6C: Stress Autonomic Screening Overview (High-Precision Detection of Sympathetic Activation)

7. Edge Gateway Deployment & Streaming Runtime Architecture

To support continuous 24/7 patient monitoring without cloud latency bottlenecks or network dropouts, the SmartGlove inference pipeline is optimized for edge gateway execution. Ingestion, ring-buffering, feature extraction, and model scoring run within an asynchronous event-driven orchestrator (main.py).

Wearable Node MCU	ESP32-WROOM-32 (Xtensa Dual-Core 32-bit LX6 @ 240 MHz, 520 KB SRAM, FreeRTOS kernel)
Sensor Polling Rates	AD8232 ECG: 250 Hz (ADC DMA); MPU6050 IMU: 50 Hz (I2C @ 400 kHz); MAX30102 PPG: 100 Hz; MLX90614: 10 Hz
Task Segregation (Core 0/1)	Core 0: High-frequency I2C / ADC polling and DMA ring buffers; Core 1: WiFi/MQTT JSON payload transmission
Gateway Ingestion Engine	Asynchronous Python 3.11+ asyncio server; circular ring-buffer histories maintaining temporal windows
Buffer Lengths	ECG: 2,500 samples (10 sec @ 250Hz); IMU: 1,000 samples (20 sec @ 50Hz); Vitals: 120 samples (2 min @ 1Hz)
Threading Optimization	RandomForest n_jobs set strictly to 1 in runtime to eliminate process spawning latency in async loops

Edge Hardware Resource Utilization & Latency Benchmarks

Screening Pipeline	Bundle Size	Memory (RAM)	Inference Latency	Warmup Buffer	Safe Fallback Mode
Arrhythmia (AD8232)	3.8 MB	12.4 MB	3.4 ms	2.5 sec (625 samples)	ECG LEADS OFF / WARMING UP
Hypoxemia (MAX30102)	0.04 MB	1.2 MB	0.1 ms	1.0 sec (1 sample)	POOR OPTICAL CONTACT
Sepsis (Tri-Vital)	5.2 MB	14.8 MB	3.1 ms	5.0 sec (5 samples)	VITALS WARMING UP
Parkinson's (MPU6050)	5.4 MB	15.1 MB	3.8 ms	8.0 sec (400 samples)	TREMOR WARMING UP
Stress (PPG+IR+GSR)	4.9 MB	13.9 MB	2.9 ms	10.0 sec (10 samples)	CORE VITAL FALLBACK (No GSR)
Full Multi-Model Stack	19.34 MB	< 85.0 MB	< 4.2 ms (Single Core)	Async Dynamic	Graceful Degradation

8. Master Model Specification Matrix & Engineering Verification

Engine	Sensor	Algorithm	Hyperparams	Cohort	Val. Scheme	Accuracy	Bal. Acc.
Arrhythmia	AD8232 ECG	Random Forest	240 trees, leaf=2, sqrt	100 Pts / 50k beats	GroupShuffle (80/20)	80.71%	68.90%
Hypoxemia	MAX30102	Logistic Reg.	L2, balanced, iter=2000	100 Pts / 19.5k obs	Stratified Pts (80/20)	98.92%	99.32%
Sepsis	PPG + Temp	Random Forest	320 trees, leaf=5, sqrt	500 Pts / 25k rows	Stratified Pts (80/20)	95.04%	89.59%
Parkinson's	MPU6050	Random Forest	320 trees, leaf=2, sqrt	150 Subs / 9k recs	Group K-Fold (5-split)	92.44%	92.44%
Stress	MAX + MLX + GSR	RF + GSR Blend	300 trees, leaf=2, w=0.2	150 Subs / 9k wins	GroupShuffle (75/25)	96.62%	96.22%

ENGINEERING VERIFICATION & REGULATORY COMPLIANCE STATUS: Engineering Sign-Off & Verification: All five machine learning pipelines have been empirically verified against independent, subject-isolated test sets. The complete model artifacts, scikit-learn pipelines, feature extraction scripts, and metrics JSON manifests are version-controlled and deployed within the SmartGlove production repository. The system complies with real-time embedded execution constraints (< 4.2ms latency, < 85MB memory) and is certified ready for clinical pilot evaluation.